ARMAN ADIL MANGAT

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PROFILE

Graduate in Artificial Intelligence (UMT Lahore, 2026) applying for an MSc in Artificial Intelligence / Computer Science. I build complete AI systems on real hardware: I led a four-member team that delivered GENIE, a fully offline voice assistant for physically impaired users, running speech recognition and a local large language model, with an on-device classifier deployed on an ESP32-S3 microcontroller. Internship work in transformer fine-tuning (94.74% on AG News). I want to deepen my foundations in machine learning and efficient, deployable AI — particularly edge inference and assistive technology.

EDUCATION

Bachelor of Science in Artificial Intelligence 2022 – 2026

University of Management and Technology (UMT), Lahore, Pakistan — Department of Artificial Intelligence

Final Year Project: GENIE — offline voice assistant and safety monitor (see below)

Relevant modules: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Knowledge Representation & Reasoning, Data Mining, Parallel & Distributed Computing, Cloud Computing, Information Security, Computer Networks

FINAL YEAR PROJECT

GENIE — Offline Voice-Controlled Smart Home & Safety Assistant 2025 – 2026

Final Year Project · Team Lead (4 members) · Supervisor: Mr. Mahmood Hussain · github.com/ArmanAdilMangat/GENIE

- Led a four-member team from proposal to final defence, building a fully offline voice assistant and safety monitor for physically impaired users on a multi-node **ESP32-S3** network — no cloud services or API keys.
- Offline host pipeline: **openWakeWord** ("Hey Genie") → **Faster-Whisper** speech-to-text → **AIML intent routing** (362 patterns) with Mistral-7B via Ollama as fallback → **Piper TTS**, served with **FastAPI**; nodes linked over Wi-Fi, ESP-NOW and USB-serial.
- Owned the sensing and face node in MicroPython: 2× MQ-2 gas, flame, DHT11, 2× MPU-6050 and 2× SW-420 sensors plus an ILI9341 animated TFT face on one cooperative loop, raising gas, flame and fall events for the emergency SMS alert path.
- Deployed on-device inference on the ESP32-S3: a **232-parameter MLP** classifier written in pure MicroPython, reaching **72.6%** held-out-session accuracy on 1,532 recorded sensor windows.
- Designed the 177.8 mm laser-cut acrylic enclosure (14 pieces, generated and audited in Python), co-owned system integration, and documented the system in a **105-page technical report**.

EXPERIENCE

AI/ML Engineering Intern (Advanced Track) — DevelopersHub Corporation Mar 2026 – Apr 2026

- Fine-tuned **BERT-base-uncased** on AG News (120K samples) with Hugging Face Trainer and fp16 on an RTX 4060: **94.74% accuracy, 94.75 macro-F1**; live demo hosted on Hugging Face Spaces.
- Built a support-ticket auto-tagger (11 classes): fine-tuned DistilBERT reached **99.82% accuracy** on 550 held-out tickets, against 98.2% for few-shot SetFit and 47.3% for zero-shot BART-MNLI.
- Built an end-to-end customer churn pipeline on IBM Telco (ColumnTransformer preprocessing, 5-fold GridSearchCV over Logistic Regression and Random Forest).

Freelance Data Specialist — Upwork 2024 – Present

- Python web scraping, B2B lead generation and AI-assisted data enrichment for international clients; managed scoping, delivery and client communication independently.

OTHER PROJECTS

Portfolio Website with RAG Chatbot 2026

React · Vite · GitHub Pages · Flask · Hugging Face Spaces · Groq

- Built and deployed a personal portfolio with an embedded retrieval-augmented chatbot (Flask backend on Hugging Face Spaces); diagnosed a production outage caused by an upstream model deprecation and migrated to a configurable model with a health-check endpoint.

EmotionCNN — Facial Emotion Classifier 2025

Python · PyTorch · CUDA

- CNN built from scratch on 64×64 grayscale faces; benchmarked CPU vs GPU (RTX 4060) training time, inference latency and accuracy.

Intrusion Detection Benchmark — CICIDS-2017 2025

Python · Scikit-learn · TensorFlow

- Compared Random Forest, CNN, MLP and a CNN-LSTM hybrid on multi-class network flows: Random Forest reached 99.93% and matched or beat every deep model; documented duplicate-flow and random-split limitations.

Smart Attendance System — IoT + Biometric 2024

C++ (Arduino) · ESP32-CAM · ESP-WROOM

- Three-member team; owned the audio-output module and communication between ESP boards.

TECHNICAL SKILLS

Programming: Python, C/C++ (Arduino-ESP32, ESP-IDF), MicroPython, SQL

Machine Learning: PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, BERT, CNN, LSTM, Pandas, NumPy

LLMs & Speech: Ollama (Mistral-7B), Faster-Whisper, openWakeWord, Piper TTS, retrieval-augmented generation

Embedded & IoT: ESP32-S3, ESP32-CAM, ESP-NOW, I2S audio, I2C sensors, PlatformIO

Tools: Git/GitHub, FastAPI, Flask, Streamlit, NVIDIA CUDA, Linux, Google Colab, Docker (basic)

CERTIFICATIONS

- IBM — Generative AI: Introduction and Applications (Coursera, 2025)
- IBM — Introduction to Artificial Intelligence (Coursera, 2025)
- NVIDIA — AI Infrastructure and Operations Fundamentals (Coursera, 2024)
- University of Michigan — Programming for Everybody (Python) (Coursera, 2024)
- DevelopersHub Corporation — AI/ML Engineering Internship Completion (Advanced Track), 2026

AWARDS

- HEC Pakistan scholarship — Coursera Plus access under the DLSEI E-Courses programme
- Certificate of Recognition from the SVP & Group Head Corporate Affairs, Allied Bank of Pakistan

LANGUAGES

English (professional working proficiency) · Urdu (native) · Punjabi (native) · Mandarin Chinese (beginner)

REFEREE

Mr. Mahmood Hussain — Final Year Project Supervisor, Department of Artificial Intelligence, University of Management and Technology, Lahore. Further references available on request.